

Metamer-set-based approach to estimating surface reflectance from camera RGB

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We present an approach to estimating the reflectance of a surface given its camera response. In this approach we first solve the general form of this problem: We calculate the set of all possible surface reflectances, called the metamer set, and then choose a member from this set. Three possibilities in choosing a single reflectance are described here. First, we assume that all reflectances are equally likely and minimize worst-case error. Second, we adopt the assumption that reflectances follow a normal probability distribution and maximize this probability. Finally, we assume that reflectances are smooth and maximize this property. The results of our experiments show that there is significant benefit from the proposed approach in terms of the accuracy of the estimation compared with that of standard estimation methods. Moreover, the present approach introduces a notion of robustness of estimates in the form of error bars. © 2006 Optical Society of America

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1. INTRODUCTION

Digital color input devices, such as cameras and scanners, typically capture three measurements of surface reflectance known as the vector RGB. Surface reflectance, however, is a continuous function of wavelength; therefore for any given RGB there may be many corresponding reflectance spectra. Since the surface reflectance is more informative and independent of the capturing conditions than the RGB, many researchers have aimed at inverting color image formation in order to recover the original reflectance function.

Traditionally, researchers have sought to render the problem solvable by imposing constraints such that reflectances and RGBs are in a one-to-one relation. For example, if reflectances reside in a three-dimensional (3D) linear subspace of the space of reflectances, then reflectance estimation is reduced to solving a 3D linear system of equations.¹ Unfortunately, a 3D linear model for reflectances is unrealistic. In general, surface reflectances are higher-dimensional² (between 5D and 8D). However, in a 5D or 8D world, it is not possible to uniquely invert the color image formation equations to recover reflectance.

While it is not possible to establish a unique correspondence between RGBs and reflectances, we show that it is possible to solve for the set of all surface reflectances that induce the same RGB. Such reflectances are called metamers,³ so we call the whole set of possible reflectances the metamer set. Metamer-set calculation consists of setting conditions to which reflectances must adhere and then solving for the reflectances that meet these constraints. In terms of these constraints, we propose that reflectances are bounded (between 0 and 100%) and that they are natural in the sense that they adhere to statistical properties found in typical reflectances. Both these constraints can be expressed in terms of linear inequalities. These inequalities delimit a convex region of reflectance space, and this is the set of plausible reflectances.

Of course, solving for the metamer set is only the first

stage of reflectance estimation. For practical purposes we need to choose a single reflectance from within this set. In this paper we consider three methods for choosing a single estimate from within the metamer set. Two are based on statistical assumptions about the distribution of surface reflectances, and one is based on the idea that surface reflectances should be “smooth” functions.^{4,5} Once chosen, however, we can describe the reflectance relative to the rest of the metamer set. Thus, in effect we can solve for reflectances with error bars. This is a very important point. In some circumstances, e.g., for saturated colors, the metamer set is small, so we can tell that the reflectance we recover is the correct answer with very small error bars. Not only is this useful for estimating possible errors through, say, a color reproduction system, it is the first recovery method we know of that has this property.

While the most important contribution of our method is its theoretical foundation, we are also able to improve actual performance of reflectance estimation. We conducted a set of experiments to compare metamer-set-based reflectance estimating to traditional methods. We found that metamer-set-based methods, with any reasonable selection criterion, improve over standard, nonmetamer-set-based methods.

In Section 2 we recapitulate the algebra needed to solve for the metamer set. In Section 3, we describe strategies to select reflectance estimates from within the metamer set. A brief description and discussion of traditional reflectance estimation algorithms is given in Section 4. In Section 5 experiments are reported. We conclude our study in Section 6.

2. METAMER SETS

Let us start by describing color image formation in mathematical terms. Assuming a so-called Mondrian⁶ world of flat, matte, and uniformly illuminated surfaces parallel to

the imaging plane, color image formation can be expressed by the simple linear equation

$$\chi_i = \int_{\omega} X_i(\lambda) E(\lambda) R(\lambda) d\lambda, \quad (1)$$

where the integral is taken over the visible range ω (usually 400 to 700 nm), and χ_i represents the i th response, $X_i(\lambda)$ the i th spectral sensitivity of the observer, $E(\lambda)$ the spectral power distribution of the illuminating light source, and $R(\lambda)$ the surface spectral reflectance. In standard digital cameras and scanners there are three recording channels (replicating the principle of three types of light sensitive cones in a human observer) resulting in a 3D response referred to as the RGB vector (because the three sensors usually measure long (red), medium (green), and short (blue) wavelengths). Equation (1) in vector form becomes

$$\underline{\chi} = \int_{\omega} \underline{X}(\lambda) E(\lambda) R(\lambda) d\lambda, \quad (2)$$

where $\underline{\chi} = [\chi_1, \chi_2, \chi_3]$ and $\underline{X}(\lambda) = [X_1(\lambda), X_2(\lambda), X_3(\lambda)]$. Note that underlining will denote a vector and boldface a matrix throughout this paper.

In reflectance estimation, given $\underline{\chi}$, $\underline{X}(\lambda)$, and $E(\lambda)$, we aim to calculate $R(\lambda)$. In effect we wish to invert Eq. (2). However, this is not easily achieved. On the left-hand side of Eq. (2) we have three degrees of freedom, yet $R(\lambda)$ has in principle infinite degrees of freedom. But, while it is not possible to uniquely recover reflectance from RGB, we have shown before that it is possible to recover the set of all possible reflectances.⁷ Because in color science two reflectances that induce the same RGB are called metamers, we first set out to recover the set of all possible reflectance estimates, the metamer set.

In order to recover the metamer set we first define a number of constraints a reflectance must satisfy. Clearly $R(\lambda)$ must be bounded in the interval $[0, 1]$: No less than no light and no more than all the light is reflected at any given wavelength. (Note that this is not true for fluorescent surfaces; however, we do not consider these here.) This property is called physical realizability and we write it as

$$\forall \lambda \in \omega, \quad 0 \leq R(\lambda) \leq 1. \quad (3)$$

A second property of surface reflectances is band-limitedness.⁸ From Nyquist's theorem we know we can represent a function by q sample points, assuming the property is represented by the first $q/2$ terms in a Fourier expansion⁹ (the sampling rate should be at least twice the maximum frequency component). Moreover, if a signal is smooth then q can be small. In color science $q = 31$ sample points (400 to 700 nm at 10 nm steps) are commonly used to represent any spectrum, and have been found to be sufficiently precise assuming the spectra are smooth.¹⁰ Let us write reflectance as the $q \times 1$ vector $\underline{r}$. Without loss of generality we are able to substitute $R(\lambda)$ with $\underline{r}$:

$$\underline{r} \approx [R(\lambda_1), R(\lambda_2), \dots, R(\lambda_q)]. \quad (4)$$

However, while 31 points may be adequate from the point of view of sampling density, they may not necessarily express the dimensionality of the reflectance spectra. Numerous studies have examined the statistics of large sets of reflectance data^{2,8,11–14} by performing principal component analysis.¹⁵ The common conclusion is that reflectances can be modeled with little error as a weighted linear sum of more than three but fewer than eight basis vectors. Let us denote the $q \times k$ matrix of k basis vectors $\mathbf{B}$ and let $\underline{\sigma}$ denote the $k \times 1$ column vector of corresponding weights. Now a reflectance $\underline{r}$ can be written as

$$\underline{r} \approx \mathbf{B} \underline{\sigma}. \quad (5)$$

Finally, we propose that surface reflectances must adhere to a naturalness constraint. On an intuitive level we say that a synthetically generated surface is natural if it is possible to create it as an additive composite of real, measured surfaces. Assume a pixel catches light from a collage of reflectances and so interprets the incoming light as the sum of the underlying physical reflectances. This collage (by our definition) forms a natural reflectance. This concept is in theory equivalent to that of Maxwell's color wheel,¹⁶ where the observation of a spinning circle induces a homogeneous color that is the combination of a set of individual colors used to paint portions of the circle.

In mathematical terms we say that a surface reflectance is natural if and only if it can be written as a convex combination of a set of representative surface reflectances. Denoting $\underline{s}_i$ the $i = 1, \dots, m$ representative surface reflectances we consider natural and w_i the m portions of these, then a reflectance $\underline{r}$ is natural if and only if

$$\exists w_i \geq 0, \quad \sum_{i=1}^m w_i = 1, \quad \underline{r} = w_1 \underline{s}_1 + \dots + w_m \underline{s}_m. \quad (6)$$

An alternative way to express naturalness, as defined in Eq. (6), is to write it as a set of linear inequalities. The set of all convex combinations of reflectances $\underline{s}_i$ can be described either by the extreme vertices of its convex hull (a minimal subset of $\underline{s}_i$ such that convex combinations of this set create all natural reflectances) or by a minimal set of linear inequalities satisfied by all $\underline{s}_i$. In three dimensions, the linear inequality representation can be thought of as corresponding to half-spaces defined by each triangular facet made up of three neighboring extreme vertices of the convex hull. Let this system consist of l linear inequalities defined by the $l \times q$ matrix $\mathbf{A}$ (the left-hand side of the system) and an $l \times 1$ column vector $\underline{b}$ (the right-hand side of the system) such that

$$\forall \underline{s}_i, \quad \mathbf{A} \underline{s}_i \leq \underline{b}. \quad (7)$$

This then translates to the following condition: A reflectance $\underline{r}$ is natural if it satisfies

$$\mathbf{A} \underline{r} \leq \underline{b} \quad (8)$$

such that the pair $[\mathbf{A}, \underline{b}]$ satisfies relation (7) for the representative reflectances $\underline{s}_i$.

In summary, we have introduced four properties of surface reflectance: physical realizability, band-limitedness (resulting in a vector representation), limited dimensionality, and naturalness. We say that $R(\lambda)$ (or $\underline{r}$) is a reason-

able reflectance if it satisfies relations (3)–(5) and (8). Let us next look at how these properties are used to solve for the metamer set.

Denoting illuminant spectral power distribution as the $q \times 1$ vector $\mathbf{e}$ and observer spectral sensitivities as the $q \times 3$ matrix $\mathbf{X}$ and combining Eqs. (2) and (5), color formation is written as

$$\chi = \mathbf{X}^T D(\mathbf{e}) \mathbf{B} \sigma, \quad (9)$$

where superscript T denotes vector transpose and $D(\cdot)$ is an operator transforming a vector into a diagonal matrix.

Assuming a fixed recording setup (observer, light source, reflectance domain), we can further simplify Eq. (9) by combining the constant part of color formation into a single $3 \times k$ matrix $\Lambda = \mathbf{X}^T D(\mathbf{e}) \mathbf{B}$ (referred to as the lighting matrix¹) expressing the linear relationship between responses and linear model weights [equivalent to reflectances through relation (5)] such that

$$\chi = \Lambda \sigma, \quad (10)$$

subject to the additional two properties we introduced in relation (3) (physical realizability) and relation (8) (naturalness), which in the framework of the linear model representation of surface reflectances are written as functions of σ . This is arrived at simply by substituting $\mathbf{B}\sigma$ for r in relations (3) and (8), which become

$$0 \leq \mathbf{B}\sigma \leq \mathbf{1}, \quad (11)$$

$$\mathbf{A}\mathbf{B}\sigma \leq b. \quad (12)$$

We also note the geometrical meaning of relations (11) and (12). Each inequality, e.g. $\mathbf{B}\sigma \leq \mathbf{1}$, describes a half-plane that divides the k -dimensional space in which σ lies into two parts, half-spaces. One half-space satisfies the inequalities, the other does not. Together, all the inequalities delimit a closed, bounded convex region in σ space. Within this region we must find the metamer set.

The general form of the solution to a linear system described in Eq. (10) depends on the properties of the matrix Λ . Assuming that the observer's spectral sensitivities (matrix $\mathbf{X}$) span a 3D subspace, and the illuminant spectral power distribution is nonzero on a reasonable interval of ω (such that the product $\mathbf{X}D(\mathbf{e})$ remains 3D), the properties of Λ depend on the basis matrix of surface reflectance $\mathbf{B}$, which consists of linearly independent, orthogonal vectors. Earlier we noted that we will consider surface reflectances to be more than three dimensional. Under these conditions, Λ represents an underdetermined set of linear equations (there are more unknowns than equations).

Now let $k, 3 < k \leq q$, be the dimension of surface reflectances, so that $\mathbf{B}$ is $q \times k$, σ is $k \times 1$, and therefore Λ is $3 \times k$. Assuming the row space of Λ is truly three dimensional, the linear system it represents has $k-3$ degrees of freedom. This means that there are $k-3$ linearly independent nonzero vectors σ that result in a response χ of zeros. So, while Λ spans three dimensions over the k -dimensional space, there necessarily exists a basis for the remaining $k-3$ dimensions, orthogonal to Λ . Let us denote this basis by the $k \times (k-3)$ matrix Ξ . It can be shown that the general form of the solution to Eq. (10) can

be written as a decomposition into a part that lies in the space of Λ , denoted σ_x , and a part that lies in the orthogonal complement, spanned by Ξ , denoted σ_0 ¹⁷:

$$\chi = \Lambda(\sigma_x + \sigma_0). \quad (13)$$

Because the response to vectors from the orthogonal complement is zero, they are referred to as metameric blacks (a black surface results in a zero response) and can be arbitrarily scaled. Thus we arrive at the general form of the solution to Eq. (10):

$$\chi = \Lambda(\sigma + \Xi \alpha), \quad (14)$$

where α is an arbitrary $(k-3) \times 1$ column vector of real scalars.

Equation (14) describes all possible metameric reflectances in the basis $\mathbf{B}$. Consequently, Eq. (14) defines a $(k-3)$ -dimensional hyperplane. Intersecting this plane with the convex hull defined in relations (11) and (12) results in a $(k-3)$ -dimensional convex set of metamers, which we denote as $\mathcal{M}(\chi)$. Any member of this set describes a σ that generates a reflectance that is a metamer.¹⁸

In Section 3 we look at how we might choose a single reflectance from $\mathcal{M}(\chi)$ as a reflectance estimate.

3. CHOOSING A SINGLE METAMER

Let us assume that all metamers are equally likely; i.e., that reflectances follow a uniform probability distribution. The metamer that best characterizes this set is the centroid, defined as the solution to

$$\hat{\sigma} = \min_{\forall \sigma_i \in \mathcal{M}(\chi)} \max_{\forall \sigma_j \in \mathcal{M}(\chi)} \|\sigma_i - \sigma_j\|. \quad (15)$$

The choice of a centroid is a choice minimizing worst-case error, and we refer to this selection as the metamer set centroid estimate MSC.

Alternatively, we can assume that reflectances do not have equal probability and choose that metamer that is most likely (an approach known as maximum likelihood estimation¹⁹). Let us denote the probability of σ by $P(\sigma)$ and conditional probability, that is the probability of σ given χ , by $P(\sigma|\chi)$; then we want to solve

$$\max_{\sigma} P(\sigma|\chi); \quad (16)$$

that is, we wish to maximize the probability of σ given the known response vector χ . Bayes's rule on conditional probability states that¹⁵

$$P(\sigma|\chi) = \frac{P(\chi|\sigma)P(\sigma)}{P(\chi)}. \quad (17)$$

Metamers by definition have the same response, so $P(\chi)$ is constant and equal to 1 and need not be taken into account, as it does not affect the search for the maximum. The conditional probability $P(\chi|\sigma)$ is constant, too, as we optimize over the metamer set and again does not affect the final conditional probability we wish to maximize. Solving for Eq. (17) is therefore reduced to solving $P(\sigma)$.

A study by Freeman and Brainard suggested and experimentally supported the idea that surface weights fol-

low a truncated normal distribution.²⁰ Here we adopt a slightly weaker assumption: The probability of a weight vector is described by a multivariate normal distribution defined as¹⁵

$$P(\underline{\sigma}) = (2\pi)^{-k/2} |\underline{\Sigma}|^{-1/2} \exp \left[-\frac{1}{2} (\underline{\sigma} - \underline{\mu})^T \underline{\Sigma}^{-1} (\underline{\sigma} - \underline{\mu}) \right], \quad (18)$$

where $\underline{\Sigma}$ is the $k \times k$ variance-covariance matrix of the distribution of weights, and $\underline{\mu}$ is the $k \times 1$ vector of the mean surface weight. Let us rewrite Eq. (18) in terms of log-likelihood (we can do this because the logarithm is a monotonic function and therefore preserves extremes):

$$\ln P(\underline{\sigma}) = \ln(2\pi)^{-k/2} + \ln |\underline{\Sigma}|^{-1/2} - \frac{1}{2} (\underline{\sigma} - \underline{\mu})^T \underline{\Sigma}^{-1} (\underline{\sigma} - \underline{\mu}). \quad (19)$$

Both $\ln(2\pi)^{-k/2}$ and $\ln |\underline{\Sigma}|^{-1/2}$ are constants and can be disregarded when maximizing $\ln P(\underline{\sigma})$. Maximizing the last term of Eq. (19) can be written as minimizing its negative:

$$\min_{\underline{\sigma}} (\underline{\sigma} - \underline{\mu})^T \underline{\Sigma}^{-1} (\underline{\sigma} - \underline{\mu}). \quad (20)$$

Now recall the decomposition of $\underline{\sigma}$ into the linear sum of a $\underline{\sigma}_x$ that results in the response and a metamer black part $\underline{\sigma}_0 = \underline{\Xi} \underline{\alpha}$. Inserting this decomposition into relation (20) and simplifying such that we leave out terms containing $\underline{\sigma}_x$ that are constant, we arrive at a minimization over $\underline{\alpha}$ in the form

$$\min_{\underline{\alpha}} 2(\underline{\sigma} - \underline{\mu})^T \underline{\Sigma}^{-1} \underline{\Xi} \underline{\alpha} + \underline{\alpha}^T \underline{\Xi}^T \underline{\Sigma}^{-1} \underline{\Xi} \underline{\alpha}. \quad (21)$$

Denoting $\underline{f} = 2(\underline{\sigma} - \underline{\mu})^T \underline{\Sigma}^{-1} \underline{\Xi}$ and $\underline{H} = \underline{\Xi}^T \underline{\Sigma}^{-1} \underline{\Xi}$ we rewrite relation (21) as

$$\min_{\underline{\alpha}} \underline{f} \underline{\alpha} + \underline{\alpha}^T \underline{H} \underline{\alpha}. \quad (22)$$

Now we see that Eq. (22) is a quadratic form of $\underline{\alpha}$, the scaling factor of the metamer black basis $\underline{\Xi}$. Furthermore, $\underline{\alpha}$ is constrained (physical realizability and naturalness) by linear inequalities. Optimizing a quadratic form subject to linear inequalities is known as a quadratic programming (QP) problem.²¹ QP problems can be solved quickly with the global best answer found. By solving this optimization, we choose the most likely metamer (assuming surface weights are drawn from a multivariate normal distribution) for a given response. We shall refer to this selection method as the metamer set maximum likelihood estimate MSL.

Another criterion of choice could be smoothness of surface reflectances. Smoothness, though not uniquely defined, has been argued on numerous occasions to be an inherent property of surface reflectances.^{4,22–24} Van Trigt^{4,25} took this intuition further and analytically derived the smoothest possible reflectance to a given response in the framework of variational calculus. We adopt his definition of smoothness, which in the discrete domain can be written

$$\sum (d\underline{r}/d\underline{i})^2. \quad (23)$$

The smaller this value, the smoother the reflectance, the smoothest reflectance being a constant, the derivative of which is zero, the minimal value of the expression in relation (23). Differentiation in the discrete domain can be performed by a premultiplication with a $q \times q$ matrix of the form⁹

$$\underline{D} = \begin{bmatrix} -1 & 1 & 0 & \dots & 0 & 0 & 0 \\ -1 & 0 & 1 & \dots & 0 & 0 & 0 \\ & & & \ddots & & & \\ 0 & 0 & 0 & \dots & -1 & 0 & 1 \\ 0 & 0 & 0 & \dots & 0 & -1 & 1 \end{bmatrix}. \quad (24)$$

Making use of the simplification that for any column vector $\underline{\Sigma}(\underline{r})^2 = \underline{r}^T \underline{r}$, we can rewrite relation (23) with the discrete derivative operator from Eq. (24) as

$$\underline{r}^T \underline{D}^T \underline{D} \underline{r}. \quad (25)$$

Since we use a linear model representation of surface reflectances $\underline{r} = \underline{B} \underline{\sigma}$, we can rewrite relation (25) as

$$[\underline{D} \underline{B} \underline{\sigma}]^T \underline{D} \underline{B} \underline{\sigma}. \quad (26)$$

Denoting $\underline{H} = \underline{B}^T \underline{D}^T \underline{D} \underline{B}$, relation (26) takes the form of relation (22) (with $\underline{f} = 0$). So, we solve for $\underline{\sigma}$ that minimizes relation (26), a quadratic form, subject to the linear inequality constraints of physical realizability and naturalness. We arrive at a QP problem formulation as in the case of the MSL estimate. Let us denote this selection as the metamer set smoothest reflectance estimate MSS.

4. COMPARATIVE METHODS

To compare the metamer-set-based methods introduced in the previous Section 3, we implement three standard approaches to reflectance estimation based on similar assumptions. Let us first implement an approach assuming surface reflectance is three dimensional, sometimes referred to as lighting matrix reflectance estimation.^{1,14,26–30} This is a special case of solving for the metamer set as outlined in Section 2. Since both the RGB vector $\underline{\chi}$ and the estimated surface reflectance weight vector $\underline{\sigma}$ are of equal dimension and related through the color image formation matrix $\underline{\Lambda}$, there is a one-to-one relation between $\underline{\chi}$ and $\underline{\sigma}$. In this model metamerism is not an issue since there are three equations of three unknowns and consequently the system is fully determined and invertible:

$$\underline{\sigma} = \underline{\Lambda}^{-1} \underline{\chi}. \quad (27)$$

This is a simple and elegant way to estimate surface reflectances; however, the assumption of 3D surface reflectance is very limiting, and furthermore this method does not guarantee that estimates will be physically realizable. We refer to this estimation algorithm as 3D reflectance estimating.

Next we implement linear least-squares reflectance estimation. The idea is to minimize the L2 Euclidean distance between a linear mapping (between RGBs and re-

flectances) and the true values, based on a set of known reflectances and their corresponding RGBs. Our implementation of this method is based on the linear model representation assuming k -dimensional surface reflectance, equivalent to the assumption we made in the metamer set derivation. So, given a training set of m reflectance weights in the $k \times m$ matrix $\mathbf{S}$ and corresponding m RGB response vectors in the $3 \times m$ matrix $\mathbf{G}$, the $k \times 3$ least-squares mapping matrix $\mathbf{T}$ minimizes the L2 Euclidean distance (error) metric

$$\|\mathbf{S} - \mathbf{T}\mathbf{G}\|_2, \quad (28)$$

and is solved for as $\mathbf{T} = \mathbf{S}\mathbf{G}^T(\mathbf{G}\mathbf{G}^T)^{-1}$ (Ref. 31). Then, for a response, the linear model weights in the basis of the training set are estimated as

$$\sigma = \mathbf{T}\chi. \quad (29)$$

We refer to the least-squares approach as the LSQ method. Clearly, in terms of the L2 Euclidean distance, this method is optimal. However, as with the 3D method, the algorithm does not guarantee that the estimated reflectance be physically possible.

Finally, we implement van Trigt's smoothest reflectance estimation^{4,25} in the discrete domain. Smoothness is defined in relation (23) and can be written in the form of relation (25) with the discrete derivative operator we introduced earlier. We minimize relation (25) subject to the equality constraint that fixes the response of the unknown reflectance to the known RGB vector and to the linear inequality constraint of physical realizability,

$$\min_r r^T \mathbf{D}^T \mathbf{D} r,$$

subject to

$$\mathbf{X}\mathbf{D}(e)r = \chi \wedge 0 \leq r \leq 1.$$

The objective function to be minimized is a quadratic form, and both the equality constraints and inequality constraints are linear. Thus van Trigt's smoothest reflectances, which he solved for analytically in the continuous domain, are solved by a QP optimization.³² We refer to this approach as the smoothest reflectance estimation method, denoted SM. This last approach does not explicitly make use of a linear model representation of surface reflectances. However, not using a linear model representation and using a linear model representation with $k=q$ dimensions is mathematically equivalent.

5. EXPERIMENTAL SETUP AND RESULTS

In this section we present two sets of experiments. First we report results of reflectance estimation from synthetic camera RGBs. In this case we calculate sets of RGB vectors χ for which we wish to estimate surface reflectances according to the color image formation Eq. (2). This corresponds to an ideal case where no noise is present, where lighting and object geometry have no effect, and color image formation is described by a set of three linear equations. In a second set of experiments we then test reflectance estimation on camera RGBs captured by a digital camera. Results from this second experiment are exposed

to both noise in the camera RGBs and discrepancies between the color image formation model and actual camera RGBs due to capture conditions.

To evaluate the performance of the reflectance estimation techniques, we use two error metrics. If r is the original reflectance and $\hat{r}$ the estimate, then the estimation error Δ is defined as

$$\Delta = \frac{1}{q} \|\hat{r} - r\|_2 = \frac{1}{q} \sqrt{(\hat{r} - r)^T (\hat{r} - r)}, \quad (30)$$

which is the L2 Euclidean distance in the space of reflectances normalized by the number of samples of the discrete representation q . The smaller this measure the closer the two reflectances (two reflectances being identical if $\Delta=0$). We shall refer to this metric as the absolute error of reflectance recovery.

Since the metamer set defines all potential estimates, it is possible to examine the worst case error Δ_T that is the distance between the selected reflectance and the furthest possible metamer:

$$\begin{aligned} \Delta_T &= \frac{1}{q} \max_{r_i \in \mathcal{M}(\chi)} \|\hat{r} - r_i\|_2 \\ &= \frac{1}{q} \max_{r_i \in \mathcal{M}(\chi)} \sqrt{(\hat{r} - r_i)^T (\hat{r} - r_i)}. \end{aligned} \quad (31)$$

Given a single camera RGB, any of the reflectances in the metamer set could be the sought original. The furthest metameric reflectance to any particular estimate therefore is an expression of the stability of each of the methods. Δ_T of course also depends on the size of the metamer set, which in turn depends on the chromaticity of the camera RGB. It is well known that responses that are close to the neutral axis (achromatic colors) have significantly larger metamer sets, whereas responses to theoretical, monochromatic stimuli would yield a single reflectance. In the former case Δ_T is expected to be of a higher magnitude, whereas in the latter case it would be zero. It has been suggested that the mathematical formulation of the metamer sets can formalize this intuition into an analytical formulation of the concept of the probability of color¹⁸; however it is beyond the scope of this paper to explore this concept in more detail.

A. Synthetic Camera RGB Experiment

At a first stage we evaluate the performance of reflectance estimation under artificial and ideal conditions. The reflectances needed to train each of the algorithms are the same as those used to test them. For methods that use a linear model basis (LSQ, 3D, MSC, MSS, and ML, the latter three being metamer-set-based), the matrix of basis vectors $\mathbf{B}$ is derived from the same reflectances used for testing the algorithms. Furthermore, the LSQ method also uses the synthetic camera RGBs for calculating the mapping matrix from RGB vectors χ to the linear model weights σ , and again the same camera RGBs were used to train the LSQ matrix. The only method that does not require any training is the SM method, which analytically solves for the smoothest possible reflectance subject only

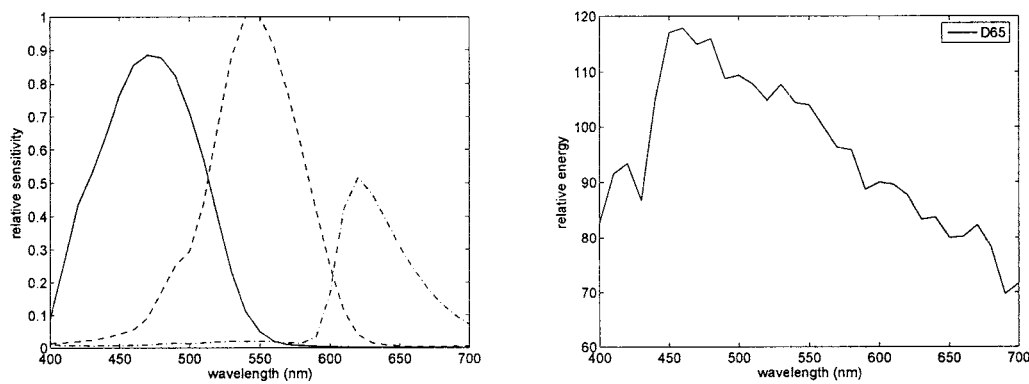


Fig. 1. Spectral sensitivities of the camera used in the synthetic data experiments and the spectral power distribution of illuminant D65.

Table 1. Reflectance Recovery Median Δ (Top Three Rows) and Δ_T (Bottom Three Rows) Statistics for the Munsell Data Set^a

Train-Test	LSQ	3D	SM	MSC	MSS	ML
All-All	0.0033	0.0033	0.0043	0.0033	0.0026	0.0028
Half-Half	0.0033	0.0034	0.0042	0.0035	0.0028	0.0029
Hull-Int.	0.0028	0.0027	0.0034	0.0028	0.0019	0.0029
All-All	0.0078	0.0078	0.0092	0.0048	0.0072	0.0068
Half-Half	0.0081	0.0081	0.0093	0.0052	0.0076	0.0073
Hull-Int.	0.0079	0.0079	0.0093	0.0053	0.0077	0.0073

^aData in boldface show the smallest and second-smallest error statistics.

Table 2. Reflectance Recovery Median Δ (Top Three Rows) and Δ_T (Bottom Three Rows) Statistics for the Natural Data Set^a

Train-Test	LSQ	3D	SM	MSC	MSS	ML
All-All	0.0080	0.0081	0.0089	0.0062	0.0068	0.0064
Half-Half	0.0076	0.0079	0.0086	0.0072	0.0072	0.0065
Hull-Int.	0.0050	0.0055	0.0065	0.0042	0.0051	0.0041
All-All	0.0139	0.0141	0.0162	0.0093	0.0150	0.0111
Half-Half	0.0122	0.0120	0.0144	0.0085	0.0134	0.0099
Hull-Int.	0.0145	0.0147	0.0167	0.0102	0.0161	0.0120

^aData in boldface show the smallest and second-smallest error statistics.

to physical realizability. This is an ideal case and only expresses the theoretical soundness of the tested approaches.

Therefore, we then evaluate the algorithms under more common conditions, where training and testing data differ. We do this in two ways. First we compare reflectance estimation methods when training on a randomly selected half of the set of reflectances and testing on the remaining half. While this is a naive way of training any algorithm, it tests robustness of performance. In a third experiment the training data are chosen more carefully. Reflectance estimation methods that, at a first stage, calculate the metamer set assume that each considered reflectance can be made up of some basic reflectances (the vertices of the convex hull of the training reflectance). Thus, in our next experiment we choose these extreme vertices of a set of data as the training reflectances, while we test on the remaining reflectances that are enclosed within this hull.

We conduct all three synthetic camera RGB experiments on two sets of data. First we use the Munsell set.³³

These are 426 generally smooth reflectances of man-made paper chips, and a 5D linear model represents spectral variation to a sufficient accuracy of over 99%. The second data set is the Natural set. This is a set of 219 natural surfaces measured by Vrhel *et al.*³⁴ containing leaves, flowers, etc. The natural data set is less smooth and therefore we use a seven-dimensional linear model in this case (also representing the data to an accuracy of over 99%). Thus for the experiments with the Munsell set **B** is 31×5 , while for the Natural set it is 31×7 .

Both data sets are evaluated with CIE illuminant D65 as the illuminating light source of the synthetic scene, recorded by a standard RGB camera. The spectral power distribution of the illuminant and the spectral sensitivity curves can be seen in Fig. 1. Tables 1 and 2 summarize the median error statistics both in terms of the actual estimation error Δ and the worst-case error measure Δ_T .

The error statistics clearly demonstrate that there is benefit in a metamer-set-based approach to reflectance estimation. For both data sets and for all combinations of

training and testing data, in terms of the median statistics, metamer-set methods reduce both actual estimation error Δ as well as worst-case error Δ_T significantly.

For the Munsell data the MSS and the ML methods show best estimation, while for the Natural data set the MSC and the ML approaches show better estimation accuracy. This is consistent with the assumptions of each of the methods. The Munsell set contains reflectances that are smooth and hence comply with the assumption of the MSS method. The Natural set is less smooth and has higher-frequency components; therefore, the more general approach of choosing the centroid of the set of reflectances is more appropriate. In both cases ML gives good results, which again is consistent and expected, since in this case the probability distribution of the linear model weights of surface reflectances of each of the sets is taken into account.

In terms of a more intuitive and perceptually meaningful CIE Lab ΔE error measure, the smallest estimation error for the Munsell data set—where $\Delta=0.0019$ —corresponds to a $\Delta E=1.59$ (for the MSS method), while the largest error, $\Delta=0.0043$, corresponds to $\Delta E=2.97$ (for the SM method). For the Natural data set, the smallest error of $\Delta=0.0042$ corresponds to $\Delta E=1.62$, while the largest error, $\Delta=0.0089$, corresponds to $\Delta E=3.69$. The relationship between Δ and CIE Lab ΔE is nonlinear, and while Δ expresses the numerical accuracy of the estimated surface reflectance vector, CIE Lab ΔE is a measure of visual difference, whereby a difference of $1\Delta E$ corresponds approximately to a just noticeable difference.

B. Raw Data Experiment

In a second experiment, images of the Macbeth ColorChecker Chart³⁵ were taken with a Hasselblad camera using a Megavision digital CCD array attached to it under three illuminants: D50, cool white fluorescent (CWF), and tungsten (A) incandescent (all data courtesy of Hewlett Packard Inc.; the kind permission and help of Paul Hubel, formerly of Hewlett Packard Laboratories, is acknowledged here). The spectral sensitivities of the camera were measured using a monochromator, and the spectral power distributions of the illuminants were measured using a white diffuser tile with a telespectroradiometer (see Fig. 2). The charts were measured with the same telespectroradiometer, resulting in the color signals of each of the 24 patches under each of the three illuminants. Reflectances of these patches were obtained by dividing the color signals by the illuminant spectral power distribution. The spectral range used here was 380 nm to 780 nm at 4 nm sampling intervals, resulting in $q=101$ samples. The raw images from the camera were processed such that responses over each of the 24 patches were averaged, resulting in 24 camera RGBs.

In this experiment the raw RGBs are taken as the input for the reflectance estimation, while the measured illuminant and spectral sensitivities define the color formation matrix. The correspondence between raw RGBs and their synthetic (theoretical) equivalents is achieved through a diagonal matrix, scaling each channel such that the absolute spectral sensitivity not accounted for when measuring the sensors is compensated. This diagonal matrix is derived such that a white patch of the raw

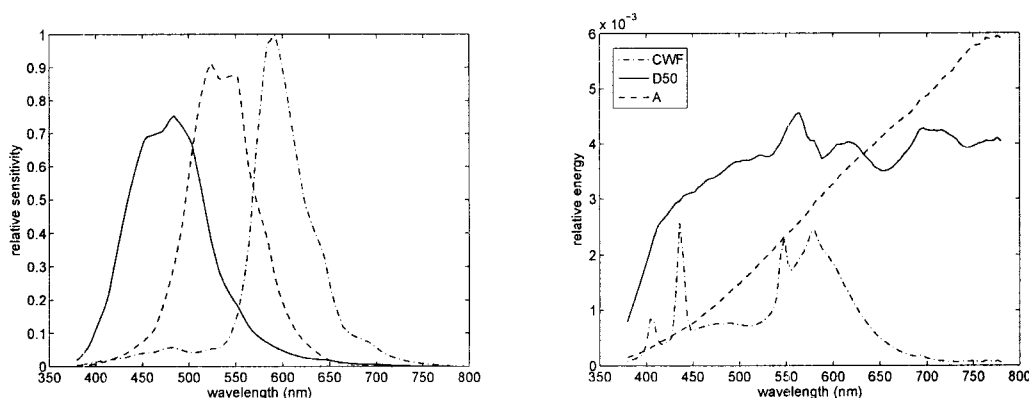


Fig. 2. Spectral sensitivities of the Hasselblad/MegaVision camera and the three illuminants used in the raw data experiments.

Table 3. Reflectance Estimating Median Δ Results (Top Three Rows) and Median Δ_T Results (Bottom Three Rows) from Raw RGB under Illuminants D50, A, and CWF^a

Illuminant	LSQ	3D	SM	MSC	MSS	ML
D50	0.0091	0.0089	0.0091	0.0088	0.0062	0.0085
A	0.0099	0.0087	0.0088	0.0094	0.0076	0.0085
CWF	0.0099	0.0101	0.0098	0.0099	0.0088	0.0092
D50	0.0137	0.0157	0.0165	0.0091	0.0127	0.0119
A	0.0146	0.0181	0.0153	0.0082	0.0118	0.0114
CWF	0.0168	0.0165	0.0176	0.0102	0.0166	0.0149

^aData in boldface show the smallest and second-smallest error statistics.

data is equal to the white patch of the synthetic counterpart. To test the algorithms using this data, each method requiring training was trained on all but one reflectance (i.e., 23 reflectances) and tested on the remaining one. This was repeated 24 times, leaving out one of the samples to test at a time.

Least-squares and 3D estimation are performed as in the synthetic camera experiments described earlier. The color formation matrix in the 3D case is derived from the three characteristic vectors of the measured reflectances, and the least-squares mapping matrix is derived from synthetic RGBs calculated from the measured reflectances. The matrix of 6 basis vectors for the metamer set estimation methods, found to be sufficient for the measured Macbeth ColorChecker, is also calculated from the measured reflectances. Table 3 summarizes the results of reflectance estimation.

The raw camera RGB estimating results in Table 3 again show the advantage of metamer-set-based reflectance estimation methods. The MSS and the ML approaches perform significantly better than any other method, reducing error over all standard approaches (LSQ, 3D, or SM). This trend holds both in terms of actual error as well as Δ_T .

6. CONCLUSIONS

In this paper we have looked at reflectance estimation. Due to the nature of color image formation, this problem is underdetermined: There is not a unique correspondence between reflectances and RGB vectors. Rather, we are confronted with the phenomenon of metamerism, whereby multiple surface reflectances record to the same RGB. In order to solve for a single reflectance estimate we first derive the general form of all metameric reflectances, the metamer set. This set defines the uncertainty inherent in the estimation process: Any reflectance in the metamer set corresponds to a physically realizable and natural reflectance representing some surface.

For practical purposes defining the set of all possible surface reflectances is not sufficient; instead a single reflectance is required. Hence we proposed a number of selection techniques to choose estimates from within the metamer set. Two methods were based on assumptions about the distribution of surface reflectances. First we assumed that all reflectances are equally likely, and in a second method we have built on previous work arguing that reflectances followed a multivariate normal distribution. The third method we developed was based on the argument that reflectances are generally smooth, and we chose the smoothest reflectance from within the metamer set.

Finally, we implemented three standard methods of reflectance estimation not based on the metamer set: linear least-squares, lighting matrix, and van Trigt's smoothest reflectance estimation. In a set of experiments we compared all methods, evaluating their performance in terms of the actual error and the maximum possible error (the furthest possible metamer). The results showed that the metamer-set-based methods consistently outperformed traditional estimation.

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